

Thomas Huang

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Skills

Languages: C#, Go, TypeScript, SQL

Frameworks & Tools: React, REST APIs, WebSockets, MCP Tooling, Stripe, Datadog

Cloud & Data: AWS Lambda, SQS, Azure OpenAI, Cloudflare, DynamoDB, MySQL, Redis, Snowflake

Practices: Event-Driven Architecture, SEO, Prompt Engineering, Context Management, Agentic Workflows, Observability

Experience

Software Engineer, Homes.com, CoStar Group – Irvine, CA

August 2023 – Present

- Built a distributed **.NET/C#** gateway leveraging persistent **WebSocket** connections to **Azure OpenAI**, with **Redis**-backed session management and cleanup to support scalable, context-aware LLM conversations
- Architected prompt-engineering workflows and **MCP** tools for LLM-powered Homes.com experiences, enabling structured tool invocation and domain-specific context retrieval
- Shipped an event-driven **Bing IndexNow** platform using **AWS Lambda**, **SQS**, **DynamoDB**, and **Snowflake**, automating property URL submissions to improve search engine crawl coverage
- Implemented **REST APIs** in **C#** powering mobile application features and user impression logging for Homes.com
- Automated marketing data pipelines with **C#** scheduled-job services integrating **MySQL** and third-party vendor APIs, supporting targeted user acquisition campaigns
- Built automated XML sitemap and CSV feed pipelines from Snowflake views that accelerated the publication of new and updated property listings to search engines and external advertising channels.

Software Engineer Intern, Imprint – Remote

January 2020 – May 2020

- Delivered the end-to-end onboarding flow for account.imprint.co, spanning landing pages, account creation, identity verification, and customer activation

Projects

Personal Website

thomhuang.com

- Built a full-stack personal website using **React**, **TypeScript**, and **Go**, featuring **JWT** cookie authentication, **Cloudflare D1/R2**-backed content storage, and containerized deployment across isolated production and test environments
- Developed a **Stripe**-powered storefront with product and inventory management, direct-to-**R2** image uploads, stock reservation, webhook-driven order fulfillment and refunds, and magic-link order viewing
- Shipped a coffee brew journal tracking roasters, origins, grinders, and brew parameters, backed by normalized lookup tables and an admin CRUD interface
- Established supporting infrastructure including request logging, health checks, idempotent schema migrations, login rate limiting, a D1-persisted token denylist, and strict CSP/security headers

Interactive Unix Shell

GitHub

- Engineered a Unix-inspired shell in **Go**, implementing process execution, built-in commands, path navigation, quoting semantics, and I/O redirection through a custom parsing and execution pipeline

Education

University of California, Los Angeles – B.S. in Applied Mathematics, Specialization in Computing 2019 – 2023